

SURIGAO, Philippines

WMO: 986530

Lat: 9.783N

Lon: 125.489E

Elev: 129

StdP: 14.63

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	74.1	74.9	70.5	112.8	77.4	71.6	117.2	78.0	17.6	80.3	15.3	80.1	4.2	180	0.347

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.7	92.8	82.0	92.0	81.9	91.2	81.7	83.9	90.3	83.1	89.8	82.5	89.4	3.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.2	168.4	88.8	81.2	162.9	88.1	80.6	159.6	87.6	48.2	90.4	47.1	90.0	46.5	89.9	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.6	11.8	10.2	71.9	96.1	2.8	2.8	69.9	98.1	68.3	99.8	66.7	101.4	64.7	103.5		
(5)			70.2	85.6	1.9	1.6	68.9	86.7	67.8	87.7	66.7	88.6	65.4	89.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.8	79.8	80.4	81.6	82.9	84.6	84.3	84.0	84.7	84.3	83.4	82.0	81.3	(6)
(7)		DBStd	2.64	2.18	2.10	2.07	1.94	1.80	2.23	2.16	2.21	2.08	2.05	2.11	2.02	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11962	924	850	981	987	1071	1028	1053	1075	1028	1034	959	972	(10)
(11)		CDD65	6487	459	430	516	537	606	578	588	610	578	569	509	507	(11)
(12)		CDH74	72467	4017	3938	5287	6051	7519	7022	7035	7536	7009	6580	5382	5091	(12)
(13)		CDH80	25636	760	869	1529	2102	3212	2923	2811	3260	2873	2423	1588	1286	(13)
(14)	Wind	WSAvg	3.8	4.9	5.1	4.9	4.2	3.4	3.0	3.4	3.1	3.4	3.3	4.0	(14)	
(15)	Precipitation	PrecAvg	143.0	24.5	17.2	14.4	9.3	5.3	5.9	6.3	6.1	6.5	10.0	15.7	19.6	(15)
(16)		PrecMax	187.7	50.6	43.5	48.8	31.8	19.7	17.7	16.9	19.8	15.3	24.7	34.7	50.4	(16)
(17)		PrecMin	105.9	3.2	0.7	1.5	1.5	0.9	1.2	0.8	0.4	0.5	1.1	1.4	7.0	(17)
(18)		PrecStd	23.3	11.4	11.2	8.2	5.9	3.5	3.3	2.9	4.1	3.6	4.3	7.3	7.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.3	88.9	90.1	91.4	93.2	93.6	93.2	93.8	93.6	92.0	90.7	89.7	(19)
MCWB			80.2	80.4	80.8	81.6	82.2	82.5	82.3	82.0	82.2	81.7	81.5	81.4	(20)	
2%		DB	86.9	87.3	88.9	90.1	91.8	92.2	91.8	92.5	92.2	90.9	89.3	88.2	(21)	
		MCWB	79.8	79.8	80.3	81.2	81.9	82.1	81.9	81.7	82.0	81.5	81.1	80.7	(22)	
5%		DB	85.5	86.2	87.8	89.1	90.9	91.0	90.7	91.4	91.0	89.8	88.0	87.0	(23)	
		MCWB	79.3	79.4	80.0	80.9	81.7	81.7	81.5	81.5	81.8	81.4	80.9	80.3	(24)	
10%		DB	84.0	84.9	86.5	87.9	89.8	89.7	89.3	90.0	89.7	88.6	86.7	85.6	(25)	
		MCWB	78.8	78.9	79.6	80.5	81.5	81.5	81.1	81.2	81.4	81.0	80.5	79.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.6	81.7	82.3	83.2	84.5	84.8	84.4	84.4	85.2	83.5	82.9	82.5	(27)
MCD B			86.8	86.9	88.1	89.1	90.4	91.1	91.0	91.1	89.9	89.7	88.4	87.7	(28)	
2%		WB	80.5	80.7	81.3	82.1	83.3	83.2	82.8	82.9	83.1	82.5	81.8	81.5	(29)	
		MCD B	85.1	86.1	87.1	88.5	89.9	90.1	89.9	90.3	90.2	89.2	87.5	86.6	(30)	
5%		WB	79.8	79.9	80.6	81.4	82.5	82.4	81.9	82.1	82.2	81.8	81.2	80.8	(31)	
		MCD B	84.1	84.9	86.4	87.6	89.5	89.5	89.0	89.5	89.7	88.2	86.5	85.7	(32)	
10%		WB	79.1	79.1	79.9	80.7	81.8	81.6	81.3	81.4	81.5	81.1	80.6	80.1	(33)	
		MCD B	83.1	83.7	85.3	86.8	88.5	88.4	87.9	88.5	88.5	87.2	85.7	84.6	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.4	8.3	9.2	9.9	10.7	10.6	10.1	10.6	10.4	9.7	8.5	8.2	(35)
MCDBR			9.9	10.4	11.3	11.5	12.0	12.4	12.1	12.4	12.3	11.6	10.9	10.3	(36)	
MCWBR			5.1	5.2	5.3	5.2	5.4	5.5	5.6	5.5	5.7	5.5	5.4	5.2	(37)	
5% WB		MCDBR	9.0	9.9	10.4	10.9	11.5	11.7	11.3	11.4	11.6	10.9	10.0	9.5	(38)	
		MCWBR	5.2	5.5	5.5	5.6	5.8	5.9	5.9	5.9	6.0	5.8	5.4	5.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.453	0.435	0.425	0.420	0.415	0.410	0.414	0.434	0.429	0.437	0.428	0.440	(40)	
taud		2.359	2.411	2.457	2.471	2.499	2.531	2.510	2.433	2.443	2.420	2.458	2.409	(41)		
Ebn at Noon		272	281	285	282	278	277	276	274	279	277	278	273	(42)		
Edh at Noon		39	38	37	36	35	33	34	37	37	37	36	37	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1158	1423	1660	1828	1794	1628	1572	1640	1621	1531	1335	1215	(44)	
(45)		RadStd	187	171	148	130	119	112	111	100	141	115	120	121	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page